

# Cover Page

Standard: BAC5004-3 Rev: (J) 31-Jan-2013

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\*\*\*\*\* PSDS GENERATED \*\*\*\*\*

**1 SCOPE**

**NOTE:** Incorporated PSDs: 6-14, 6-18, 6-19  
Canceled PSDs: None

This specification and [BAC5004](#) establish the requirements for installation and inspection of non-fluid-tight, permanent-type, straight-shank blind fasteners as specified on the applicable Engineering drawings. See [BAC5047](#) for fastener installation in fluid tight applications. The Engineering requirements for installation and inspection of fasteners in fiber/resin composites are found in [BAC5063](#). Refer to [BAC001PREF](#) for guidance on use of Boeing process specifications and Boeing process specification departures.

**2 CLASSIFICATION**

Not applicable to this specification.

**3 REFERENCES**

The current issue of the following documents shall be considered a part of this specification to the extent herein indicated.

- |                           |                                                                         |
|---------------------------|-------------------------------------------------------------------------|
| <a href="#">BAC5004</a>   | - Installation of Permanent Fasteners                                   |
| <a href="#">BAC5004-1</a> | - Solid Rivet Installation                                              |
| <a href="#">BAC5034</a>   | - Temporary Protection of Production Materials, Parts and Assemblies    |
| <a href="#">BAC5047</a>   | - Fluid Tight Fastener Installation                                     |
| <a href="#">BAC5063</a>   | - Fastener Installation in Composite Structures                         |
| <a href="#">BAC5117</a>   | - Electrical Bonding                                                    |
| <a href="#">BSS7022</a>   | - Procedures and Equipment for Measurement of Fastener Holes            |
| <a href="#">BSS7286</a>   | - Statistical Process Control of Designated Engineering Characteristics |
| D-590                     | - Boeing Part Standards Manual                                          |

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**BLIND FASTENER INSTALLATION****BAC5004-3****BOEING PROCESS SPECIFICATION**

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CAGE CODE 81205

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**5 MATERIALS CONTROL**

Not applicable to this specification.

**6 FACILITIES CONTROL**

Not applicable to this specification.

**7 DEFINITIONS**

Engineering  
Drawing

- The collection of product definition data used to disclose, directly or by reference, through pictorial or textual presentations, or combinations of both, the physical and functional end product requirements and configuration of an item. The term may be used regardless of the actual medium or method used for its depiction. A drawing may be computer-aided, manually produced, digitally defined within a dataset and plotted, or digitally defined within a dataset and not plotted.

**8 MANUFACTURING CONTROL****8.1 HOLES**

- a. Hole sizes for blind rivets shall be in accordance with [Table I](#).
- b. Hole sizes for blind bolts shall be in accordance with [Table II](#), [Table III](#), and [Table IV](#).
- c. Hole sizes for [BACB30AU](#) blind bolts shall be as specified on Engineering drawings.

**TABLE I - HOLE SIZES FOR BLIND RIVETS**

| HOLE DIAMETERS FOR <a href="#">NAS1738</a> , <a href="#">NAS1739</a> , <a href="#">BACR15FP</a> , <a href="#">BACR15FR</a> , <a href="#">BACR15GJ</a> , <a href="#">BACR15GK</a> AND CR6253 |                      |       |                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-------|-----------------------------------------------------------------------|
| RIVET DIAMETER<br>NOMINAL (INCH)                                                                                                                                                            | HOLE DIAMETER (INCH) |       | DRILL DIAMETER (DRILL SIZE) <a href="#">FL 1</a> <a href="#">FL 2</a> |
|                                                                                                                                                                                             | MIN                  | MAX   |                                                                       |
| 1/8                                                                                                                                                                                         | 0.143                | 0.146 | 0.144 (27)                                                            |
| 5/32                                                                                                                                                                                        | 0.176                | 0.180 | 0.177 (16)                                                            |
| 3/16                                                                                                                                                                                        | 0.205                | 0.209 | 0.2055 (5)                                                            |
| HOLE DIAMETER FOR <a href="#">BACR15GR</a>                                                                                                                                                  |                      |       |                                                                       |
| RIVET DIAMETER<br>NOMINAL (INCH)                                                                                                                                                            | HOLE DIAMETER (INCH) |       | DRILL DIAMETER (DRILL SIZE) <a href="#">FL 1</a> <a href="#">FL 2</a> |
|                                                                                                                                                                                             | MIN                  | MAX   |                                                                       |
| 3/32                                                                                                                                                                                        | 0.098                | 0.101 | 0.0980 (40)                                                           |
| ALL OTHER BLIND RIVETS                                                                                                                                                                      |                      |       |                                                                       |
| RIVET DIAMETER<br>NOMINAL (INCH)                                                                                                                                                            | HOLE DIAMETER (INCH) |       | DRILL DIAMETER (DRILL SIZE) <a href="#">FL 1</a> <a href="#">FL 2</a> |
|                                                                                                                                                                                             | MIN                  | MAX   |                                                                       |
| 3/32                                                                                                                                                                                        | 0.097                | 0.101 | 0.0980 (40)                                                           |
| 1/8                                                                                                                                                                                         | 0.128                | 0.132 | 0.1285 (30)                                                           |
| 5/32                                                                                                                                                                                        | 0.160                | 0.164 | 0.1610 (20)                                                           |
| 3/16                                                                                                                                                                                        | 0.192                | 0.196 | 0.1920                                                                |
| 1/4                                                                                                                                                                                         | 0.256                | 0.261 | 0.2590                                                                |

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## 8.1 HOLES (Continued)

**FL 1** The target hole diameter shall be as small as practical. The target hold diameter shall fall within the specification limits. The target hole diameter shall allow free insertion of the rivet. Scraping of the rivet or the hole is not acceptable.

**FL 2** Recommended drill size.

**TABLE II - HOLE SIZES FOR BLIND BOLTS [BACB30AY](#), [BACB30CC](#),  
AND [BACB30JG](#)**

| NOMINAL SIZE (INCH) | HOLE DIAMETER LIMITS (INCH) |         |
|---------------------|-----------------------------|---------|
|                     | MINIMUM                     | MAXIMUM |
| 5/32                | 0.165                       | 0.168   |
| 3/16                | 0.199                       | 0.202   |
| 1/4                 | 0.260                       | 0.263   |
| 5/16                | 0.312                       | 0.315   |
| 3/8                 | 0.375                       | 0.378   |
| 7/16                | 0.437                       | 0.441   |

**TABLE III - HOLE SIZES FOR BLIND BOLTS [MS90353](#), [MS90354](#),  
[MS21140](#), [MS21141](#)  
[BACB30VX](#) AND [BACB30VW](#)**

| NOMINAL SIZE (INCH) | HOLE DIAMETER LIMITS (INCH) |         |
|---------------------|-----------------------------|---------|
|                     | MINIMUM                     | MAXIMUM |
| 5/32                | 0.164                       | 0.167   |
| 3/16                | 0.199                       | 0.202   |
| 1/4                 | 0.260                       | 0.263   |
| 5/16                | 0.312                       | 0.315   |
| 3/8                 | 0.374                       | 0.377   |
| 7/16                | 0.437                       | 0.441   |
| 1/2                 | 0.500                       | 0.504   |

**TABLE IV - OVERSIZE FASTENERS AND HOLE SIZES FOR OVERSIZE BLIND BOLTS**

| ORIGINAL FASTENER                                                  | OVERSIZE FASTENER                                                    |                                | OVERSIZE     | HOLE SIZES FOR OVERSIZE REPLACEMENT FASTENER (INCH) |                |                |                |                |
|--------------------------------------------------------------------|----------------------------------------------------------------------|--------------------------------|--------------|-----------------------------------------------------|----------------|----------------|----------------|----------------|
|                                                                    |                                                                      |                                |              | 5/32                                                | 3/16           | 1/4            | 5/16           | 3/8            |
|                                                                    |                                                                      |                                |              | OVERSIZE ORIGINAL HOLE SIZE                         |                |                |                |                |
|                                                                    | FL 3                                                                 | FL 2                           |              | 0.164<br>0.167                                      | 0.199<br>0.202 | 0.260<br>0.263 | 0.312<br>0.315 | 0.374<br>0.377 |
| MS21140-()()<br>MS21140-()()P                                      | <a href="#">BACB30VJ</a> ()C()<br><a href="#">BACB30VJ</a> ()CD()    | OB100-EU()()<br>OB100-EU()()CD | 1/64<br>FL 1 |                                                     |                |                |                |                |
| MS21141-()()<br>MS21141-()()P                                      | <a href="#">BACB30VH</a> ()C()<br><a href="#">BACB30VH</a> ()CD()    | OBP-EU()() OBP-EU()()<br>( )CD |              |                                                     |                |                |                |                |
| MS90353-()()<br>MS90354-()()                                       | <a href="#">BACB30VJ</a> ()-()<br><a href="#">BACB30VH</a> ()-()     | OB100-T()() OBP-T()()<br>( )CD |              | 0.180<br>0.183                                      | 0.215<br>0.218 | 0.276<br>0.279 | 0.328<br>0.331 | 0.390<br>0.393 |
| <a href="#">BACB30VX</a> ()()()<br><a href="#">BACB30VW</a> ()()() | <a href="#">BACB30VX</a> ()() )X<br><a href="#">BACB30VW</a> ()() )X | N/A<br>N/A                     |              |                                                     |                |                |                |                |
|                                                                    |                                                                      |                                |              |                                                     |                |                |                |                |

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8.1 HOLES (Continued)

**FL 1** Quality Assurance verification required.

**FL 2** For sizes 5/32 and 3/16 the OB in the part number indicates double action tooling, for single action tooling the prefix shall be OSB.

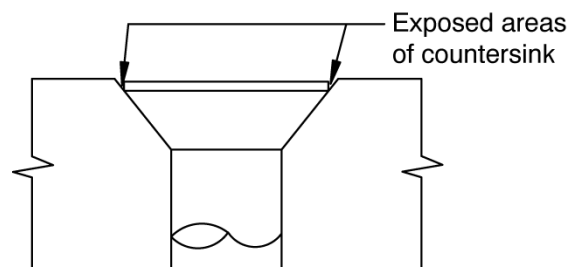
**FL 3** [BACB30VJ](#) and [BACB30VH](#): Size 5/32 and 3/16 require double action tooling unless the part number has an S suffix to the length dash number, which indicates single action tooling. Sizes 1/4 and larger are only available in single action type.

[BACB30VW](#): Sizes 5/32 and 3/16 require double action tooling unless the part number has an S or U suffix to the length dash number, which indicates single action tooling. Sizes 1/4 and larger are only available in single action type.

[BACB30VX](#): Sizes 5/32 and 3/16 require double action tooling unless the part number has a D or U suffix to the length dash number, which indicates single action tooling. Sizes 1/4 and larger are only available in single action type.

8.2 INSTALLATION

- a. Remove all temporary coatings from mating and faying surfaces prior to assembly, except that the temporary coating may remain in place on the surface that the fastener will enter, provided:
  - (1) Countersunk head fasteners can be installed through temporary protective coating that conforms to [BAC5034](#) if rivet flushness requirements are met.
  - (2) Temporary protective coating remaining after countersinking may remain under the manufactured head of countersunk fasteners provided head gapping allowances are met.
- b. Manufacturing tolerances on the heads of flush-head fasteners may result in partially exposed countersinks (see [Figure 1](#)). Exposed areas around the fastener head are acceptable if the fastener tolerances, hole tolerances, and flushness requirements are met.



**FIGURE 1 - PARTIALLY EXPOSED COUNTERSINK**

- c. Install the fastener in the hole and use an appropriate tool to pull the stem of the fastener and complete the installation.

**NOTE:** When it is practical, the parts should be clamped prior to installation of the fastener to help assure a tight joint.

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8.3 GRIP LENGTH

- a. Grip length tables are contained in the applicable part standards (See D-590).
- b. The use of washers is controlled in [Section 8.5](#) and shall be considered when grip lengths are selected.
- c. Fastener grip length may be adjusted up or down one grip length to accommodate variations in stack-up tolerances. For the purposes of this specification the stack up thickness is the total thickness of the materials being fastened together, measured at the fastener hole and excluding any gaps between the materials.

8.4 CORROSION PROTECTION

Unless otherwise specified on the Engineering drawing, all alloy steel blind bolts shall have the entire head (including stem break-off area) protected in accordance with the finish requirement on the Engineering drawing.

8.5 USE OF WASHERS

- a. Blind bolts

The use of washers for the grip length adjustment of protruding-head bolts is limited to one washer with a maximum nominal thickness of 0.0625 inch. When used, the washer shall be located under the manufactured head. Unless otherwise specified, use a compatible washer material and finish as noted in [Table V](#).

**TABLE V - WASHER SELECTION FOR USE WITH BLIND FASTENERS**

| STRUCTURE AGAINST WHICH THE WASHER BEARS | WASHER MATERIAL | WASHER FINISH                   |
|------------------------------------------|-----------------|---------------------------------|
| Aluminum                                 | Aluminum        | Anodized <a href="#">FL 1</a>   |
|                                          | CRES            | Cadmium <a href="#">FL 2</a>    |
| Alloy Steel                              | CRES            | Cadmium <a href="#">FL 2</a>    |
| Titanium/CRES                            | CRES            | Passivated <a href="#">FL 3</a> |

**FL 1** Example: [BACW10BN](#)( )AC

**FL 2** Example: [BACW10BP](#)( )CD, [BACW10BN](#)( )SC

**FL 3** Example: [BACW10BP](#)( )ACU, [BACW10BN](#)( )UC

- b. Blind rivets in thermoplastics See [BAC5063](#).

8.6 PART SEPARATION

- a. After installation, all blind fasteners shall have no measurable gap at the fastener shank except for [BACR15DH](#) and except for blind fasteners used for the sole purpose of attaching nut plates and gang channels.
- b. Part separation at the shank shall not exceed 0.020 inch after installing parts with [BACR15DH](#) nonstructural shoulder rivet

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8.6 PART SEPARATION (Continued)

- c. Blind fasteners used for the sole purpose of attaching nut plates and gang channels shall have a maximum gap less than or equal to 0.004 inch at the fastener shank.

8.7 FLUSHNESS

## a. Flushness Limits

Flushness limits for flush-head fasteners shall be + 0.010/- 0.005 inch unless otherwise noted on the Engineering drawing.

## b. Blind Rivets

The lock rings (lock collars) and/or stems of [NAS1399](#), [NAS1739](#), [BACR15FP](#), [BACR15GE](#) and [BACR15GJ](#) rivets may be shaved a maximum of 0.010 inch to meet flushness requirements. Quality Assurance inspection and approval of lock ring and stem position prior to shaving is required. Shaving of the fastener head is not allowed.

## c. Blind Bolts

(1) The protruding threaded bolt portion of [BACB30AY](#) fasteners may be shaved to meet the flushness requirements. Quality Assurance inspection and approval of bolt position prior to shaving is required. Shaving of fastener head is not allowed.

(2) The lock ring (mechanical lock) and/or stems (serrated pins) of [MS90353](#), [MS21140](#) and [BACB30VX](#) fasteners may be shaved to meet flushness requirements. Quality Assurance inspection and approval of lock ring and stem position prior to shaving is required. Shaving of the fastener head is not allowed.

8.8 REWORK

- a. Document rework as required by the applicable quality assurance provisions.

- b. [MS21140](#), [MS21141](#), [MS90353](#), [MS90354](#), [BACB30VX](#), and [BACB30VW](#) fasteners may be replaced with oversize fasteners in accordance with the limitations shown in [Table IV](#).

9 **MAINTENANCE CONTROL**9.1 GAGES

- a. Tools and gages that are used to verify the requirements of [Section 11](#) must comply with the requirements specified in [AS9100](#).
- b. Protrusion gages, feeler stock, or grip gauges which are out of tolerance shall not be used.

10 **QUALITY CONTROL**

- a. Quality Assurance shall assure that the requirements of this specification are met by monitoring the process and examining the end-items in accordance with established Quality Assurance provisions.

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**QUALITY CONTROL (Continued)**

- b. The inspection reliability requirement (IRR) values stated in [Table VI](#) define the level of verification required to demonstrate conformance. Sample in accordance with [BSS7286](#). Sample size in accordance with [Table VIII](#).
- c. Verify that only appropriately maintained installation tools and certified inspection equipment are used for performing this process.
- d. Hole diameter measurement procedures and equipment must comply with [BSS7022](#) where variable measurements are used for inspection. Attribute measurements that are taken with GO and NO-GO gages are permitted. The part of the gage used to evaluate the maximum hole diameter must be able to enter an elongated hole.

**TABLE VI - INSPECTION RELIABILITY REQUIREMENTS (IRR)**

| CHARACTERISTIC                                       | IRR (PERCENT) |
|------------------------------------------------------|---------------|
| 1. Holes                                             | 98            |
| a. Diameter                                          |               |
| b. Surface defects                                   |               |
| c. Angularity                                        |               |
| d. Countersinking and dimpling                       | 97            |
| 2. Flushness                                         |               |
| 3. Gaps and part separation                          |               |
| 4. Locking ring and stem position after installation |               |

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## QUALITY CONTROL (Continued)

TABLE VII - SAMPLE SIZE REQUIREMENTS

| LOT SIZE     | INITIAL RELIABILITY REQUIREMENT |     |
|--------------|---------------------------------|-----|
|              | 97                              | 98  |
|              | SAMPLE SIZE                     |     |
| Up to 25     | All                             | All |
| 26 to 50     | 19                              | 25  |
| 51 to 102    | 20                              | 29  |
| 103 to 106   | 21                              | 30  |
| 107 to 111   | 21                              | 30  |
| 112 to 116   | 21                              | 30  |
| 117 to 123   | 21                              | 30  |
| 124 to 130   | 21                              | 31  |
| 131 to 140   | 21                              | 31  |
| 141 to 155   | 21                              | 31  |
| 156 to 200   | 22                              | 32  |
| 201 to 214   | 22                              | 32  |
| 215 to 233   | 22                              | 32  |
| 234 to 270   | 22                              | 32  |
| 271 to 323   | 22                              | 33  |
| 324 to 403   | 22                              | 33  |
| 404 to 502   | 22                              | 33  |
| 503 to 633   | 22                              | 33  |
| 634 to 917   | 22                              | 34  |
| 918 to 1555  | 23                              | 34  |
| 1556 to 5000 | 23                              | 34  |
| 5001 and Up  | 23                              | 34  |

11

## REQUIREMENTS

11.1

HOLES

- a. Hole diameters shall be in accordance with [Section 8.1](#) or the Engineering drawing, as applicable.
- b. Surface defects, angularity, countersinking and dimpling shall be in accordance with the requirements of [BAC5004](#).

11.2

FLUSHNESS

Flushness limits for flush head fasteners shall be + 0.010/- 0.005 inch unless otherwise specified on the Engineering drawing.

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11.3 GAPS AND PART SEPARATION

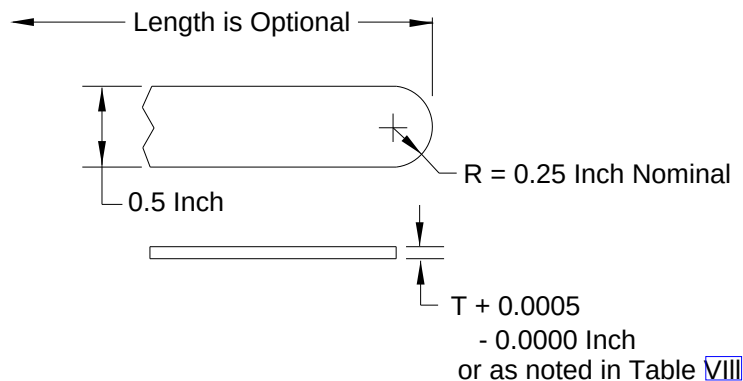
## 11.3.1 GAP EVALUATION

Gaps shall be evaluated as follows:

- a The heads of protruding-head blind fasteners shall seat with no gap in at least one location on the periphery of the head. The remaining gap shall be such that the feeler stock specified in [Table VIII](#) and [Figure 2](#) can not be inserted under the fastener head in accordance with [Figure 3](#).
- b The heads of all flush-head blind fasteners shall seat with no gap in at least one location on the periphery of the head. The remaining gap shall be such that a 0.003 inch-thick feeler stock in accordance with [Figure 2](#) cannot be inserted between the fastener head and the countersink in accordance with [Figure 4](#).
- c For gaps on tube assemblies see [BAC5004-1](#).

**TABLE VIII - FEELER STOCK THICKNESS**

| NOMINAL FASTENER<br>DIAMETER (INCHES) | FEELER STOCK<br>THICKNESS (INCH)<br>$T \pm 0.0005$<br>0.0000 |
|---------------------------------------|--------------------------------------------------------------|
| 3/32                                  | 0.0030                                                       |
| 1/8                                   | 0.0030                                                       |
| 5/32                                  | 0.0040                                                       |
| 3/16                                  | 0.0050                                                       |
| 1/4                                   | 0.0060                                                       |
| 5/16                                  | 0.0070                                                       |
| 3/8                                   | 0.0080                                                       |
| 7/16                                  | 0.0090                                                       |
| 1/2                                   | 0.0100                                                       |

**FIGURE 2 - GAP INSPECTION FEELER STOCK****BAC5004-3**

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## 11.3.1 GAP EVALUATION (Continued)

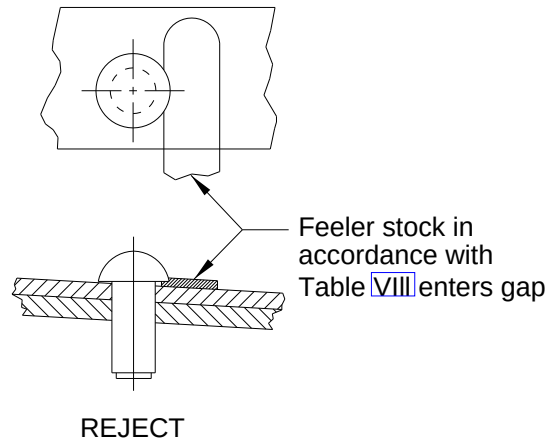


FIGURE 3 - PROTRUDING HEAD GAP CHECKING METHOD

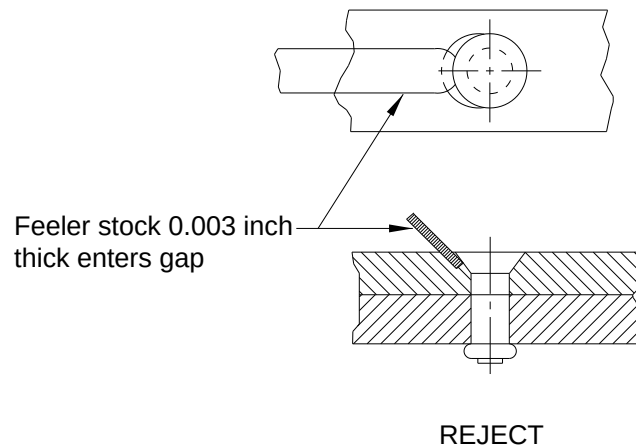


FIGURE 4 - FLUSH HEAD GAP CHECKING METHOD

## 11.3.2 PART SEPARATION

Part separation shall be in accordance with [Section 8.6](#).

11.4 LOCK RING AND STEM POSITION AFTER INSTALLATION

Where practical, the blind side of the installation shall be inspected to ensure expanded sleeves are in contact with the structure. Cracks in sleeves are not allowed.

## 11.4.1 INSPECTION OF BLIND RIVETS AFTER INSTALLATION

- a. [NAS1398](#), [NAS1399](#), [NAS1738](#), [NAS1739](#), and [BACR15GE](#) lock ring and stem positions shall be inspected in accordance with [Table IX](#) and [Figure 5](#).

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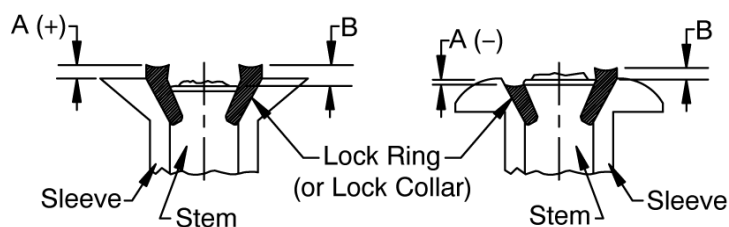
## 11.4.1

## INSPECTION OF BLIND RIVETS AFTER INSTALLATION (Continued)

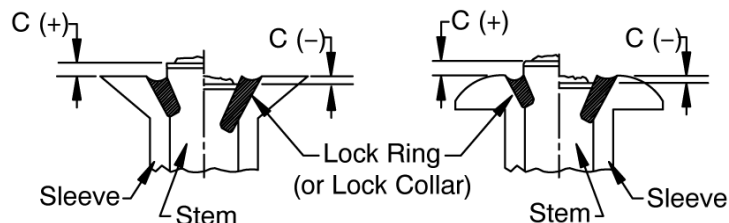
**TABLE IX - LOCK RING AND STEM INSPECTION REQUIREMENTS FOR  
[NAS1398](#), [NAS1399](#), [NAS1738](#), [NAS1739](#) AND [BACR15GE](#) BLIND RIVETS  
 (SEE Figure 5 )**

| NOMINAL RIVET<br>DIAMETER (INCH) | A (INCH)    | B (INCH) | C (INCH) |         |
|----------------------------------|-------------|----------|----------|---------|
| 3/32                             | $\pm 0.015$ | + 0.015  | + 0.020  | - 0.010 |
| 1/8                              | $\pm 0.020$ | + 0.015  | + 0.020  | - 0.010 |
| 5/32                             | $\pm 0.020$ | + 0.020  | + 0.020  | - 0.010 |
| 3/16                             | $\pm 0.020$ | + 0.025  | + 0.020  | - 0.010 |
| 1/4                              | $\pm 0.020$ | + 0.030  | + 0.020  | - 0.010 |

## LOCK RING PROTRUSION LIMITS



## STEM PROTRUSION LIMITS



**FIGURE 5 - LOCK RING AND STEM PROTRUSION LIMITS**

A = Maximum allowable distance of lock ring above (+) or below (-) top surface of fastener head.

B = Maximum allowable distance of lock ring above top of land on stem.

C = Maximum allowable distance of top of land on stem above (+) or below (-) top surface of fastener head.

- b. [BACR15FP](#), [BACR15FR](#), [BACR15GJ](#), [BACR15GK](#), and CR6253 lock collar and stem positions shall be inspected in accordance with [Table X](#) and [Figure 5](#) and [Figure 6](#).

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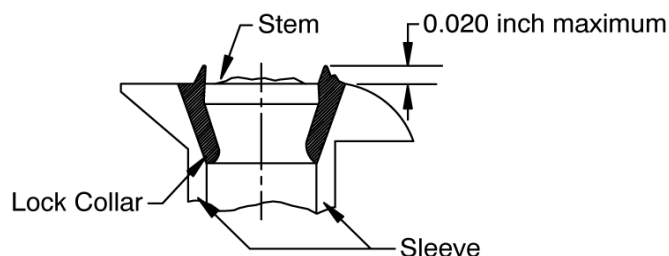
\*\*\*\*\* PSDS GENERATED \*\*\*\*\*

## 11.4.1 INSPECTION OF BLIND RIVETS AFTER INSTALLATION (Continued)

**TABLE X - STEM AND LOCK COLLAR INSPECTION REQUIREMENTS FOR [BACR15FP](#) AND [BACR15FR](#), [BACR15GJ](#), [BACR15GK](#) AND CR6253 BLIND RIVETS (SEE Figure 5 )**

| NOMINAL RIVET DIAMETER (INCH) | A (INCH) <a href="#">FL 1</a> | B (INCH) | C (INCH) |         |
|-------------------------------|-------------------------------|----------|----------|---------|
| 1/8                           | +0.005 -0.000                 | ---      | +0.010   | - 0.015 |
| 5/32                          | +0.005 -0.000                 | ---      | +0.010   | - 0.020 |
| 3/16                          | +0.005 -0.000                 | ---      | +0.010   | -0.020  |

**FL 1** 0.020-inch lock collar flash permissible (see [Figure 6](#))



**FIGURE 6 - LOCK COLLAR FLASH POSITION ([BACR15FP](#), [BACR15FR](#), [BACR15GJ](#), [BACR15GK](#) AND CR6253)**

- c. [BACR15GR](#) stem position shall be inspected in accordance with the following:

The maximum allowable distance of top of stem above (+) or below (-) top surface of the fastener head shall be +0.000/-0.025. The top of the [BACR15GR](#) stem is the highest point on the stem fracture surface.

**NOTE:** [BACR15GR](#) blind rivets have no lock ring/collar.

## 11.4.2 INSPECTION OF BLIND BOLTS AFTER INSTALLATION

- a. [BACB30AY](#), [BACB30CC](#), and [BACB30JG](#) bolt break-off positions shall be inspected in accordance with [Table XI](#) and [Figure 7](#).

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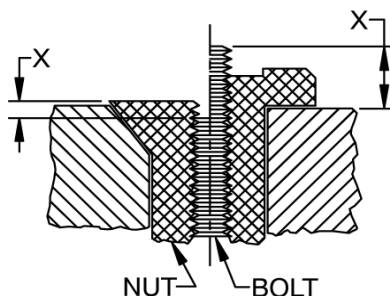
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## 11.4.2

## INSPECTION OF BLIND BOLTS AFTER INSTALLATION (Continued)

**TABLE XI - BOLT BREAK-OFF LIMITS FOR [BACB30AY](#),  
[BACB30CC](#), AND [BACB30JG](#) BLIND BOLTS**

| BREAK-OFF LIMITS (X) (INCH)                                                                                                                                                                                                                                   |                          |                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------------|
| NOMINAL DIAMETER<br>(INCH)                                                                                                                                                                                                                                    | FLUSH-HEAD BOLTS<br>FL 1 | PROTRUDING- HEAD<br>BOLTS FL 2 |
| 5/32                                                                                                                                                                                                                                                          | + 0.020<br>- 0.068       | + 0.098<br>- 0.000             |
| 3/16                                                                                                                                                                                                                                                          | + 0.015<br>- 0.073       | + 0.108<br>- 0.000             |
| 1/4                                                                                                                                                                                                                                                           | + 0.010<br>- 0.078       | + 0.145<br>+ 0.037             |
| 5/16                                                                                                                                                                                                                                                          | + 0.010<br>- 0.083       | + 0.156<br>+ 0.033             |
| 3/8                                                                                                                                                                                                                                                           | + 0.010<br>- 0.093       | + 0.162<br>+ 0.039             |
| 7/16                                                                                                                                                                                                                                                          | - - -<br>- - -           | + 0.220<br>+ 0.125             |
| <p><b>FL 1</b>      + denotes bolt protrusion above fastener head.<br/>              - denotes bolt is below fastener head.</p> <p><b>FL 2</b>      + denotes bolt protrusion above skin surface.<br/>              - denotes bolt is below skin surface.</p> |                          |                                |



**FIGURE 7 - BOLT BREAK-OFF LIMITS**

- b. [MS90353](#), [MS90354](#), [MS21140](#), [MS21141](#), [BACB30VX](#) and [BACB30VW](#) lock ring (mechanical lock) and stem (serrated pin) positions shall be inspected in accordance with [Table XII](#) and [Figure 5](#).

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## 11.4.2

## INSPECTION OF BLIND BOLTS AFTER INSTALLATION (Continued)

**TABLE XII - LOCK RING AND STEM INSPECTION REQUIREMENTS FOR [MS90353](#), [MS90354](#), [MS21140](#), [MS21141](#), [BACB30VX](#) and [BACB30VW](#) BLIND BOLTS (SEE [Figure 5](#))**

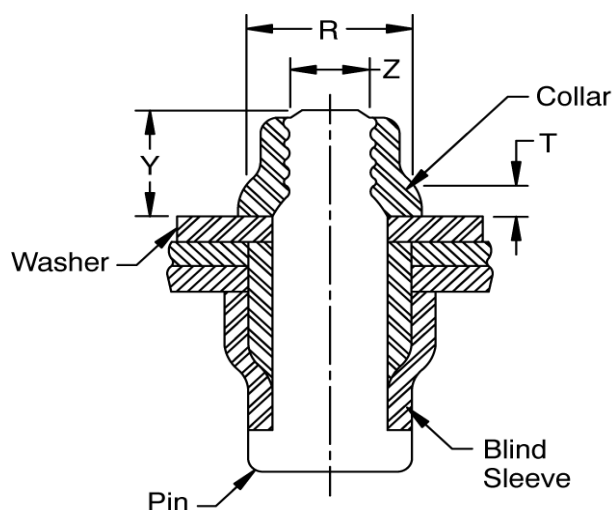
| NOMINAL RIVET DIAMETER (INCH) | A (INCH) | B (INCH) | C (INCH) |
|-------------------------------|----------|----------|----------|
| 5/32                          | ± 0.017  | ---      | ± 0.010  |
| 3/16                          | ± 0.022  | ---      | ± 0.012  |
| 1/4                           | ± 0.029  | ---      | ± 0.015  |
| 5/16                          | ± 0.037  | ---      | ± 0.019  |
| 3/8                           | ± 0.045  | ---      | ± 0.023  |
| 7/16                          | ± 0.050  | ---      | ± 0.027  |
| 1/2                           | ± 0.056  | ---      | ± 0.031  |

- c. [BACB30AU](#) pin protrusion and collar swage shall be inspected in accordance with [Table XIII](#) and [Figure 8](#).

**TABLE XIII - PIN PROTRUSION AND COLLAR SWAGE INSPECTION DIMENSIONS FOR [BACB30AU](#) BLIND BOLTS**

| NOMINAL DIAMETER | BOLT PART NO. | Y (INCH)<br><a href="#">FL 1</a> | Z (INCH) | R (INCH) | T (INCH) | GAGE NO.              |
|------------------|---------------|----------------------------------|----------|----------|----------|-----------------------|
| 1/4              | BACB30AU8     | 0.280<br>0.208                   | 0.164    | 0.303    | 0.039    | HG85-10<br>ST8711DS-6 |
| 5/16             | BACB30AU10    | 0.374<br>0.295                   | 0.224    | 0.400    | 0.038    | HG85-2<br>ST8711DS-8  |

**FL 1** When collars are installed against a sloping surface, the pin protrusion shall be checked with the plane of the protrusion gage oriented at 90 degrees to the slope of the surface.



R – Swaged collar Reference Diameter

T – Maximum height of R above sheet or washer.

Y – Pin protrusion – height of Z diameter above sheet or washer.

Z – Reference Diameter locates measuring probe for Y.

**FIGURE 8 - INSPECTION OF [BACB30AU](#)**

\*\*\*\*\* PSDS GENERATED \*\*\*\*\*

12            **TEST METHODS**

Not applicable to this specification.

13            **QUALIFICATION**

Not applicable to this specification.